

AWS IAM Least Privilege Lambda and S3 Demo

Demonstrated here is a secure S3 access control using AWS Lambda execution roles, custom IAM policies, and CloudWatch logging of events

Project Workflow

1. A Lambda function attempts to access an object **“AWS Simple File.txt”** in an S3 bucket
2. The initial request fails due to missing IAM permissions
3. A custom least-privilege IAM policy is created and attached to the lambda function
4. The Lambda successfully retrieves the object
5. CloudWatch logs then capture the execution details

Figure 1: Amazon S3 bucket containing test objects used for Lambda access validation. S3 demo bucket named **“monicac-demo-s3”** that is used to store test objects and validate the lambda access permissions.

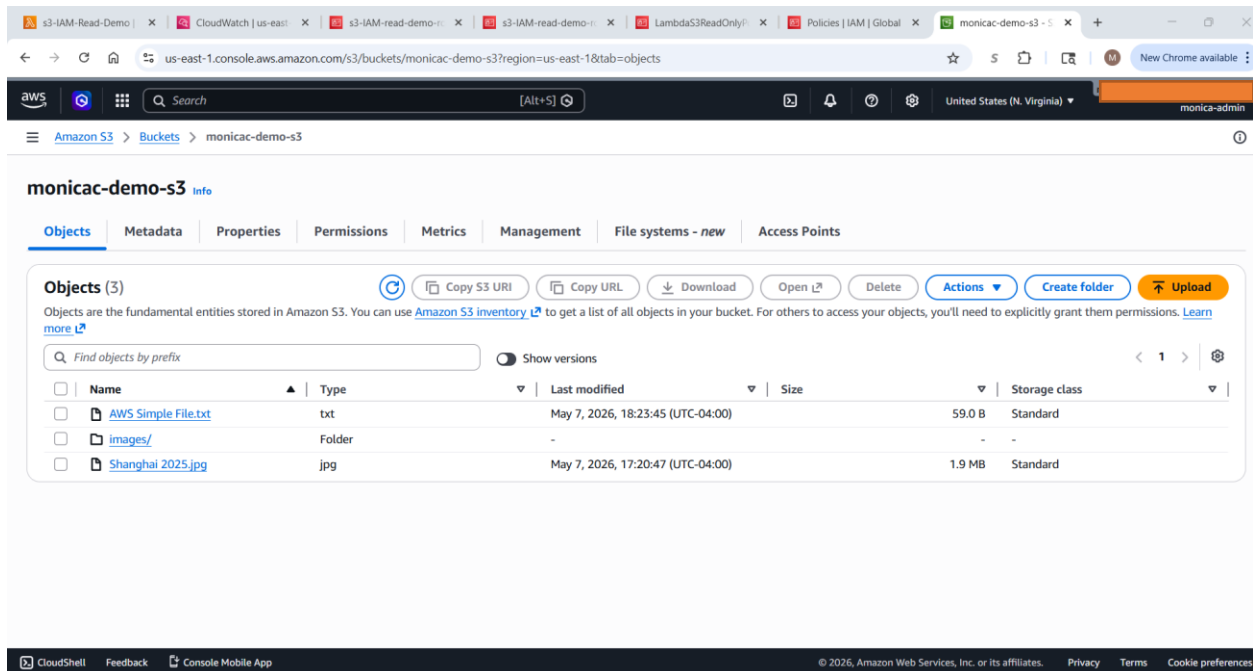
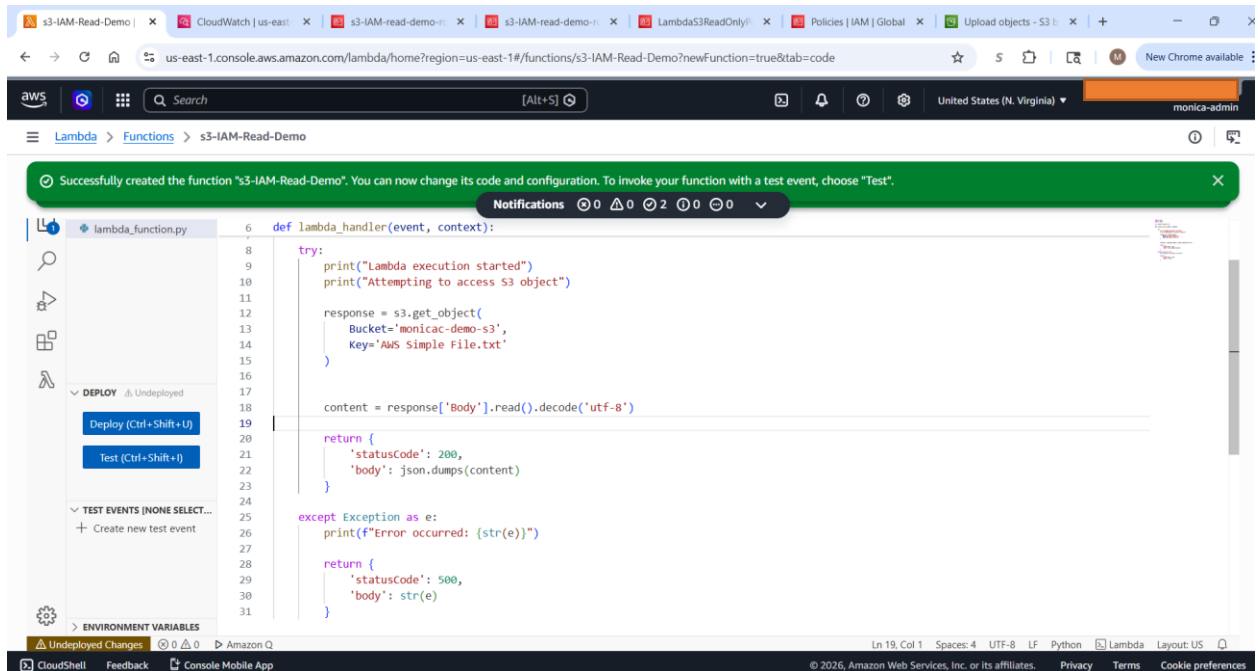


Figure 2 — AWS Lambda function configured to retrieve objects from an S3 bucket using boto3.

The Lambda function attempted to access object in the S3 bucket called **“AWS Simple File.txt”** with the following Python script



The screenshot shows the AWS Lambda console for the function 's3-IAM-Read-Demo'. A green notification bar at the top states: 'Successfully created the function "s3-IAM-Read-Demo". You can now change its code and configuration. To invoke your function with a test event, choose "Test".' Below this, the code editor displays the following Python code:

```
1  def lambda_handler(event, context):
2
3      try:
4          print("Lambda execution started")
5          print("Attempting to access S3 object")
6
7          response = s3.get_object(
8              Bucket='moniac-demo-s3',
9              Key='AWS Simple File.txt'
10         )
11
12         content = response['Body'].read().decode('utf-8')
13
14         return {
15             'statusCode': 200,
16             'body': json.dumps(content)
17         }
18
19     except Exception as e:
20         print(f"Error occurred: {str(e)}")
21
22         return {
23             'statusCode': 500,
24             'body': str(e)
25         }
```

The left sidebar shows the 'Deploy' button (Ctrl+Shift+U) and 'Test' button (Ctrl+Shift+I). The bottom status bar indicates the code is in Python and the function is in the 'us-east-1' region.

Figure 3 — Initial Lambda execution failure caused by missing `s3:GetObject` permissions. The following images demonstrated IAM deny by default behavior before the least-privilege access configured. The Lambda function returned a “Access Denied” message after attempting to read the S3 object called “AWS Simple File.txt”.

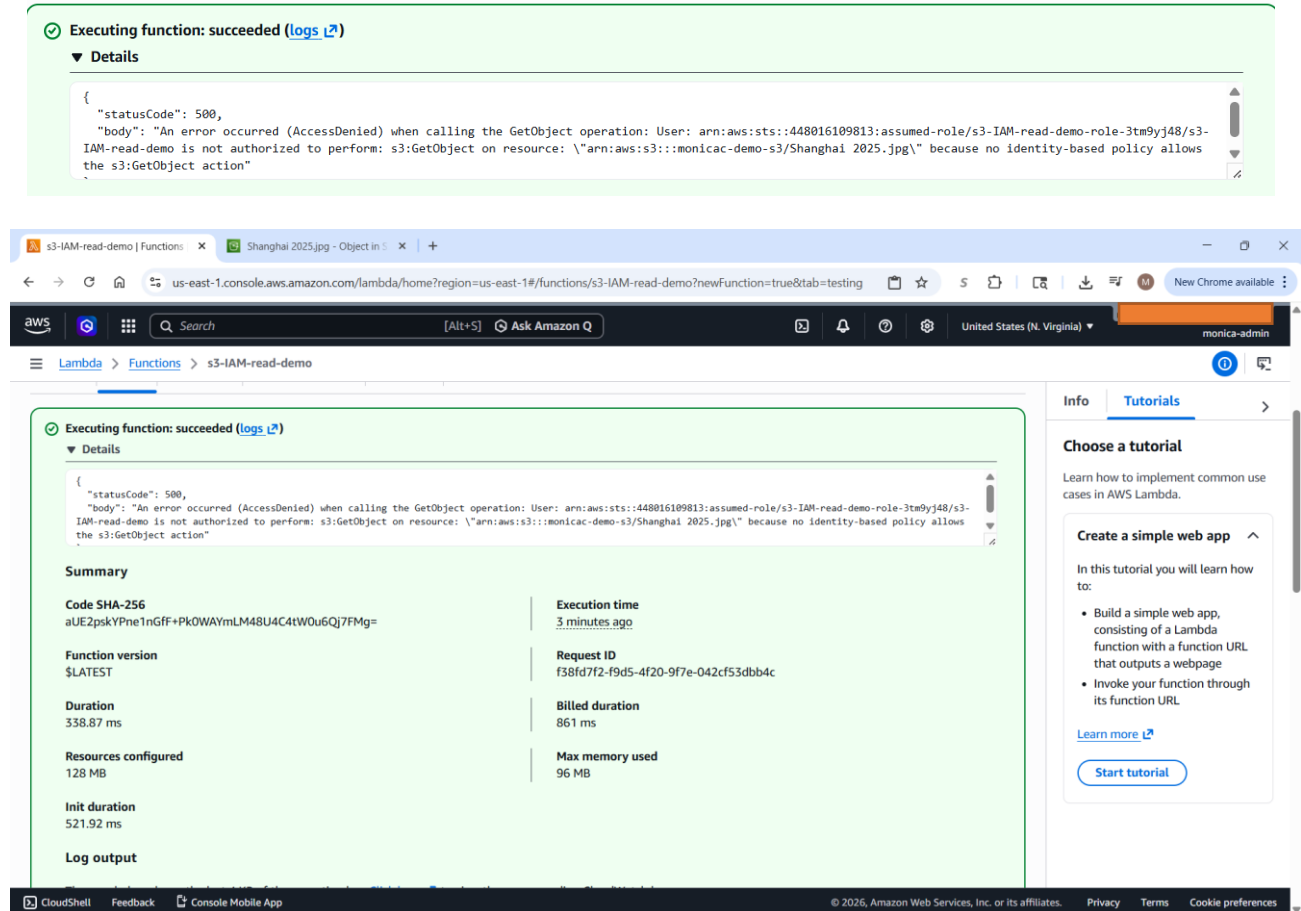


Figure 4 — Custom least-privilege IAM policy granting scoped s3:GetObject access to a single S3 bucket.

A custom IAM policy was created but restricted the Lambda function to access only the approved S3 bucket “monicac-demo-s3”.

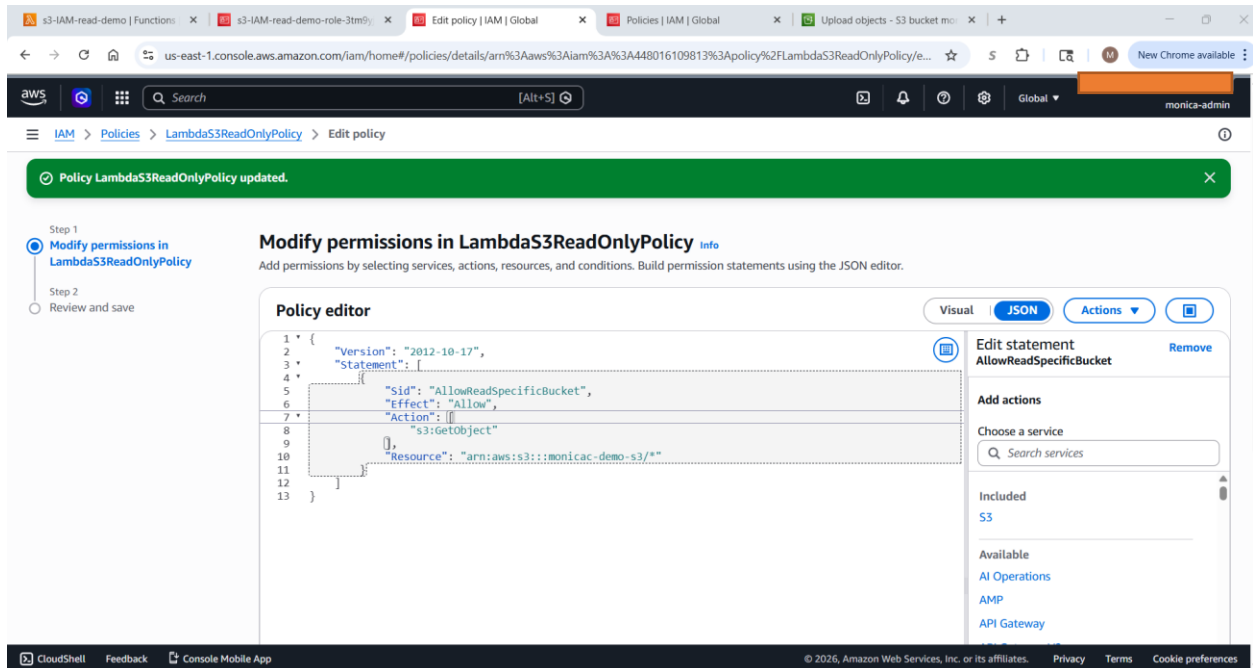


Figure 5 — Lambda execution role with attached custom IAM policy.
Here the new policy was attached to the Lambda function.

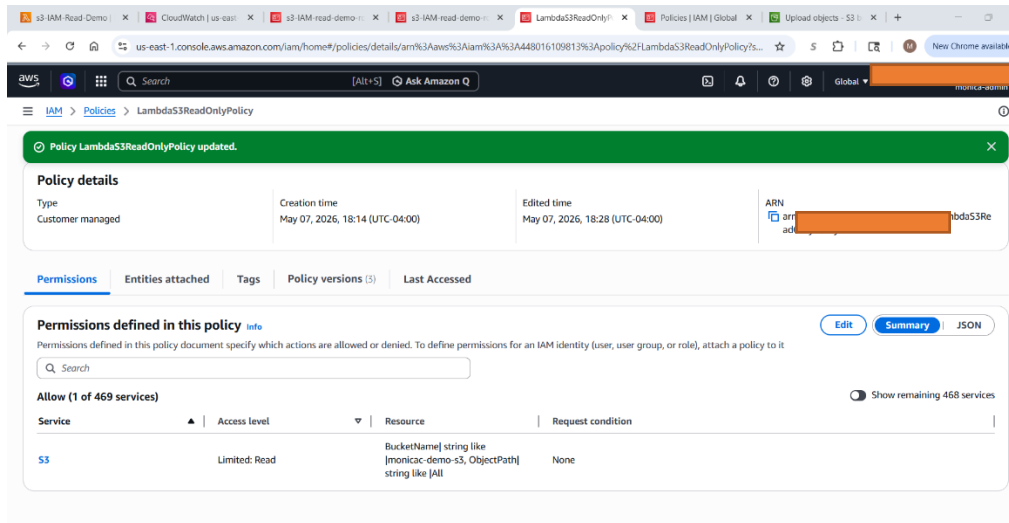


Figure 6 — Successful Lambda execution after applying least-privilege IAM permissions.
The Lambda function was able to successfully retrieve the S3 object "AWS Simple File.txt" after applying the permission. The message stating "Hello there. This is my text simple file for my AWS demos."



s3-IAM-read-demo | Functions

s3-IAM-read-demo-role-3tm9y

Edit policy | IAM | Global

Policies | IAM | Global

Upload objects - S3 bucket mor

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/s3-IAM-read-demo?subtab=permissions&tab=testing

New Chrome available

aws

Search

[Alt+S]

United States (N. Virginia)

monica-admin

Lambda > Functions > s3-IAM-read-demo

Successfully updated the function "s3-IAM-read-demo".

Executing function: succeeded (logs)

Details

```
{
  "statusCode": 200,
  "body": "\"Hello there. This is my test simple file for my AWS demos. \""
}
```

Summary

Code SHA-256

sQNDOcRG2ZSBgo99zV93H2qv4Z7uUVYgQ3y6FhO+07U=

Execution time

40 seconds ago

Function version

\$LATEST

Request ID

884aaf78-ca0e-48a7-8342-6b0eeebd167

Duration

224.33 ms

Billed duration

225 ms

Resources configured

128 MB

Max memory used

97 MB

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

START RequestId: 884aaf78-ca0e-48a7-8342-6b0eeebd167 Version: \$LATEST

CloudShell Feedback Console Mobile App

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CloudWatch Monitoring

CloudWatch Logs were used to monitor Lambda execution flow, troubleshoot permission issues, and validate successful object retrieval operations.

The screenshot displays the AWS CloudWatch console interface. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#logsV2:log-groups/log-group/%2Faws%2FLambda%2Fs3-IAM-read-demo`. The console header includes the AWS logo, a search bar, and the user's name 'monica-admin'.

The main content area is titled **/aws/lambda/s3-IAM-read-demo**. It features several action buttons: **Actions**, **View in Logs Insights**, **Start tailing**, and **Search log group**.

The **Log group details** section is expanded, showing the following information:

- Log class:** Standard
- ARN:** `arn:aws:logs:us-east-1:448016109813:log-group:/aws/lambda/s3-IAM-read-demo:*`
- Creation time:** 57 minutes ago
- Retention:** Never expire
- Stored bytes:** -
- Metric filters:** 0
- Subscription filters:** 0
- Contributor Insights rules:** -
- KMS key ID:** -
- Deletion protection:** Off
- Data protection:** -
- Sensitive data count:** -
- Custom field indexes:** [Configure](#)
- Transformer:** [Configure](#)
- Anomaly detection:** [Configure](#)
- Bearer token authentication:** Off

Below the details, there is a horizontal menu with tabs: **Log streams**, **Tags**, **Data protection**, **Anomaly detection**, **Metric filters**, **Subscription filters**, **Contributor Insights**, **Field indexes**, and **Transformer**.

The **Log streams (5)** section is visible, with a note: *By default, we only load the most recent log streams.* It includes buttons for **Delete**, **Create log stream**, and **Search all log streams**.

The footer of the console shows links for **CloudShell**, **Feedback**, and **Console Mobile App**, along with copyright information: © 2026, Amazon Web Services, Inc. or its affiliates. It also includes links for **Privacy**, **Terms**, and **Cookie preferences**.